MIL-L-48768(AR)
AMENDMENT 3
14 November 1983
SUPERSEDING
AMENDMENT 2
17 December 1979

MILITARY SPECIFICATION

LITHIUM RESERVE BATTERY ELECTROLYTE

This Amendment forms a part of Military Specification MIL-L-48768(AR), dated 19 June 75, and is approved for use by the US Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Delete "(PA)" wherever it appears and substitute "(AR)".

Delete the Preamble and substitute the following:

"This specification is approved for use by the US Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense."

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3.2 Properties. Delete table I in its entirety and substitute the following:

"Table I

Property	Requirement	Applicable Paragraph
Color	Clear & Colorless to Pale Yellow	4.4.3.1
Specific Gravity at 25.0 ± 1.0 C	1.2850 to 1.3150	4.4.3.3
Purity		
Water, ppm, max.	150	4.4.3.5
Methanol, ppm, max.	300	4.4.3.6
Formic Acid, ppm, max.	100	4.4.3.6
Dimethyl Ether, ppm, max.	100	4.4.3.6
Halogens:		
Inorganic Halogens, ppm, max.	30	4.4.3.11.1
Organic Halogens, ppm, max.	30	4.4.3.11.2

FSC 6810

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Table I (Continued)

Property	Requirement	Applicable Paragraph
Total Impurities, ppm, max.	500	4.4.3.6
pH at 25° ± 2°C	2.8 - 6.5	
Diffusion Current, Microamps	20	4.4.3.9
Specific Conductance mmho/cm at 25° ± 1.0°C	39 to 46	4.4.3.10"

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3.2 First article testing. Correct paragraph number "3.2" to "3.3".

PAGE 4

4.3.1 Submission. Delete "1 gallon" and substitute "1 liter".

4.4.2 Testing. At the end of the sentence add the following: "(at a relative humidity not to exceed 4% at ambient conditions)".

* 4.4.2.1:

Sampling. Delete "2 liters" and substitute "500 mL".

Delete the table shown in its entirety and substitute the following:

"Table II. Classification of defects

Test/Examination	Requirement Paragraph	Defect Classification
Color	3.2	Major
Specific gravity	3.2	Major
Purity	3.2	Major
Total impurities	3.2	Major
Diffusion current	3.2	Major
Specific conductance	3.2	Major"

4.4.3.2 Insoluble matter. Delete this paragraph in its entirety.

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* 4.4.3.3.1 Equipment. Delete in its entirety and substitute the following:

"Equipment and materials.

a. Two calibrated pycnometers fitted with rubber stoppers.
b. 25 cc syringe fitted with needle.
c. Controlled temperature chamber: 25°C ± 1.0°C.
d. Solution to be tested."

4.4.3.3.2 Procedure. Delete temperature "25 ± 0.1°C" and substitute "25 ± 1.0°C".

4.4.3.4 Viscosity. Delete this paragraph in its entirety.

4.4.3.6:

Methanol. Delete "wood" and substitute "wool".

Delete "20" and substitute "30".

4.4.3.7 Refractive index. Delete this paragraph in its entirety.

4.4.3.8 pH. Delete in its entirety and substitute the following:

"4.4.3.8 pH. Determine the pH of the electrolyte in accordance with method ASTM-E70 at 25 ± 2°C except that electrolyte shall be diluted in the ratio of one part electrolyte to nine parts of boiled distilled water. Measure pH within 3 minutes following preparation of solution."

4.4.3.10 Specific conductance. Delete "0.1" and substitute "1.0".

4.4.3.10.1:

Equipment. Delete "10,000 Hertz" and substitute "1K - 10K Hertz".

In procedure, third sentence after "separate readings" and before "in ohms", insert "per cell". Also, add the following: "Record three (3) separate readings per cell."

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* 4.4.3.11 Delete in its entirety and substitute the following:

"4.4.3.11 Halogen determination.

4.4.3.11.1 Inorganic halogen. The procedure outlined below is the Turbidimetric analytical technique. The colorimetric or nephelometric standard analytical techniques may also be used.

a. Equipment for Turbidimetric technique:

(1) Turbidimeter (Hach Chemical Model 2100A or equivalent).

(2) Reagents: Sodium Chloride Solution (5 μg/ml)
Silver Nitrate Solution (5%)
Nitric Acid (concentrated)

b. Procedure for turbidimetric method:

(1) Make up standard halogen solutions as follows:

Place 0, 1, 2 and 3 ml of 5 μg/ml sodium chloride solution respectively into four 50 ml flasks, each of which contains 1 ml of concentrated HNO_3 and 1 ml of 5% $AgNO_3$. Fill up to the mark with water, shake well, let them age in a dark place for 10 minutes, and measure turbidity. Plot a standard curve of halogen concentration against turbidity.

(2) In a beaker, dissolve 0.5 ml of methyl formate in 15 ml of water.

(3) In a separate container, make up a solution of diluted nitric acid using 27 ml nitric acid and 9 ml water.

(4) Transfer the methyl formate solution to a 50 ml volumetric flask and rinse the beaker with three 11 ml portions of the nitric acid solution. Add the rinsings to the volumetric flask. The total volume of solution will now be approximately 48 ml.

(5) Measure turbidity, and use this as the blank value. The turbidity of this solution is normally as low as that of deionized water.

(6) Add 1 ml of 5% $AgNO_3$ to the flask, and fill to the 50 ml mark with water.

(7) Shake well and let stand in a dark place for 10 minutes.

(8) Measure turbidity and determine the total halogen content by using the standard curve.

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4.4.3.11.2 Organic halogen.

a. Equipment:

(1) Gas chromatograph (Hewlett-Packard 5840A or equivalent).

(2) Detector - Flame Ionization (FID) or Electron Capture (ECD).

(3) Install glass injection port liner between septum and column inlet.

(4) Column - Silicone Gum Capillary (Hewlett-Packard SE54 or equivalent).

NOTE: Stainless steel column, OV-101 (6 meters length, 1/8 inch diameter) may also be used.

b. Gas chromatograph conditions:

(1) Detector temperature: 290°C.

(2) Column temperature: 40°C.

(3) Carrier gas: Helium at 60 ml/min.

(4) Injection temperature: 290°C.

(5) Sample size: 1 µl.

c. Procedure:

(1) Inject 1 µl of electrolyte.

(2) Compare the peak area with the known standards for quantitative analysis, and identify any unknown peaks.

(3) Change glass injection port liner prior to each sample injection.

4.4.3.11.3 Organic Halogen. Alternate procedure.

a. Equipment:

(1) Gas chromatograph (Hewlett-Packard 5840A or equivalent).

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(2) Detector - Flame Ionization (FID) or Electron Capture (ECD).

(3) Column - Silicone Gun Capillary (Hewlett-Packard SE54 or equivalent).

NOTE: Stainless steel column, OV-101 (6 meters length, 1/8 inch diameter) may also be used.

b. Gas chromatograph conditions. The GC conditions shall be the same as those specified in 4.4.3.11.2, Section b.

c. Procedure:

(1) Distill approximately 1.5 ml of electrolyte to separate the volatile components from the non-volatile components.

(2) Inject 1 µl of distilled volatile component.

(3) Compare the peak area with the known standards for quantitative analysis, and identify any unknown peaks."

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* 6.2 Delete "ATTN: SARPA-QA-A-P, Picatinny Arsenal" from address given and substitute "ATTN: DRSMC-QAR-R(D)". ".

6.5 Handling and storage. After "room temperature" add "or lower".

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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